

MAISHA RAHMAN CHOWDHURY

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EDUCATION

Iowa State University, Ames, IA

Aug 2022 – May 2026

Bachelor of Science in Computer Science, Minor in Applied AI

GPA: 3.72/4.00

- Magna Cum Laude, Dean's List

UNDERGRADUATE RESEARCH

Adaptive Agentic Scheduling for LLM Inference, Research Assistant | Dr. Asad W. Malik May 2026 – Present

- Extending iGenOrch for adaptive OS-level orchestration of concurrent LLM inference.
- Applying LinUCB contextual bandits to learn resource-management policies from runtime telemetry.
- Evaluating mean- and P90-based scheduling with dynamic token-budget adaptation.
- Exploring predictive agentic scheduling for proactive resource allocation under workload contention.
- Integrating LangGraph for telemetry collection, policy decisions, scheduling, and reward updates.

Summer iEDGE REU | Mississippi State University | Undergraduate Research Fellow May 2025 – Aug 2025

- Developed a Linux kernel-aware scheduler on Raspberry Pi 5 to prioritize concurrent LLM tasks.
- Deployed TinyLLaMA and DeepSeek models with automated startup and monitoring tools.
- Collected and analyzed latency, CPU, and memory metrics using Python.
- Improved inference efficiency through dynamic scheduling and C-based system-level controls.

Code-LLM Interpretability Research | Iowa State University

Jun 2024 – Aug 2024

- Researched the interpretability of code-trained large language models.
- Built a Tkinter annotation tool in Python to compare GPT-generated and human labels.
- Identified error patterns in LLM-generated labels, including insufficient context and imprecision.

PUBLICATIONS

Akram, F., Chowdhury, M. R., Dunnne, G., Williams, K., Love, K., Malik, A. W., & Khan, S. U.

“iGenOrch: Intelligent Orchestration Framework for Multi-Model LLM Inference on Edge Platforms.”

ACM Southeast Conference (ACMSE), 2026.

PRESENTATIONS

Undergraduate Research Showcase, Mississippi State University

July 31, 2025

TECHNICAL SKILLS

Programming: Python, C, C++, Java, SQL, JavaScript/TypeScript

Systems: Linux, Kernel Scheduling, Bash/Unix, Raspberry Pi

AI/ML: LLM Evaluation, Edge Inference, Neural Networks LangGraph, Machine Learning Algorithms

Frameworks & Tools: React, Node.js, Firebase, Git, Conda, Tkinter, LangGraph, LM Studio

WORK EXPERIENCE

Throwly – Full-Stack Startup Application

Jan 2026 – May 2026

- Implemented impression-based content ranking to dynamically sort items by engagement and popularity.
- Integrated third-party identity verification to enhance platform security and user authenticity.
- Built modular, type-safe frontend components in React/TypeScript, and backend development in Node.js.

Teaching Assistant, Department of Computer Science, Iowa State University

Aug 2024 – Dec 2024

- Taught Python laboratory sessions and assisted students during office hours with course concepts and assignments.
- Graded student work and collaborated with faculty and teaching assistants to develop exam grading rubrics.

Math Grader, Department of Mathematics, Iowa State University

Aug 2023 – May 2024

- Graded assignments and exams while providing accurate,
- Held office hours to give consistent feedback to support students' mathematical problem-solving skills.

Tutor, Academic Success Center, Iowa State University

Jan 2023 – Apr 2023

- Provided individualized mathematics and computer science tutoring,
- Adapted instruction to students' learning styles and exam-preparation needs.